

Zhiyuan Cheng

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EDUCATION

Ph.D. in Biological Science, *Francis Crick Institute*, UK 2026.09-2030.06
● Degree Provider: University College London (UCL)

BSc. (Hons) Biochemistry with Professional Experience, *The University of Manchester*, UK 2022.09-2026.06
● Grade: **First Class**

RESEARCH EXPERIENCE

Undergraduate Researcher, Gladstone Institutes, USA, Jennifer A. Doudna Lab 2024.12-2025.08
Project 1: Selective elimination of TP53 mutant cancer cells using SuCas12a2-based transcript-activated chromatin shredding

- Programmed SuCas12a2's collateral cleavage activity to kill cancer cells bearing undruggable mutations.
- Engineered SuCas12a2 variants via rational design to enhance catalytic activity.
- Designed and optimized lipid nanoparticle (LNP) protocols for in vitro and in vivo mRNA delivery.
- Developed a computational pipeline for automated image segmentation and statistical validation to quantify cell-killing efficiency.

Project 2: Expanding the Cas12a2 Toolkit: Ortholog Characterization to Inform TP53-Targeted Cancer Therapies

- Performed dropout screens to define PFS requirements across Cas12a2 orthologs.
- Assessed cleavage activity and gRNA tolerance to identify variants with distinct PAM/PFS preferences.

For more information about my contribution during this period, please click the link to read my full placement report:
https://drive.google.com/file/d/1pUOCGwhLw58nVrzlcO5V_uavf0wQHd-N/view

Undergraduate Researcher, University of California, San Francisco, USA, Guo Huang Lab 2024.09-2024.12

Project 1: Investigating the Role of Taco1 in Cardiac Development (CRISPRi):

- Designed and cloned CRISPRi guide RNAs targeting Taco1 under a cardiac-specific (cTnT) promoter.
- Packaged constructs into high-titer AAV9 ($>5 \times 10^{12}$ vg/ μ L) and delivered into neonatal (P1) mouse hearts.
- Performed immunostaining to assess cardiomyocyte proliferation and transfection efficiency.
- Quantified cardiomyocyte proliferation, ploidy, and nucleation in treated vs. control samples.

Project 2: Investigating the effects of Nuclear Gene Expression Asymmetry on Heart Regeneration:

- Applied RNA Scope to examine *myl3* expression in paired nuclei of bi-nucleated cardiomyocytes.

For more information about my contribution during this period, please click the link to see more:

https://docs.google.com/presentation/d/1yrw3_YZbvAaRgDdFHevPnqYFf2lmZEPu/edit?usp=sharing&oid=108417928867411578345&rtopf=true&sd=true

SKILLS

Molecular Biology: Gel electrophoresis, plasmid extraction, PCR, qPCR, cloning, site-directed mutagenesis, transfection, transformation, RNA extraction, Western blotting, in vitro transcription, and CRISPR interference.

Cellular Biology: Cell culture, immunofluorescence staining, cryosectioning, lentivirus and AAV packaging, iPSC maintenance, Lipid nanoparticle (LNP) making, and RNA in situ hybridization

Imaging, data analysis, and bioinformatics: Fiji macros programming, Ilastik training, and use. Live cell imaging analysis with IncuCyte Image quantification and statistical analysis using GraphPad Prism 10, BLAST, and sequence alignment.

OTHER WORK EXPERIENCE

IELTS Teaching Assistant, Shenyang New Oriental School, China (During Summer holiday) 2022.09-2024.09
Peer Tutor: Mentored five students who failed exams to pass in their resits. 2023.08-2025.08

ORAL PRESENTATION

Exploiting Cas12a2 *trans*-DNA cleavage for cancer “genome shredding” *Exit Talk, Doudna Lab*, Aug 2025

PUBLICATIONS UNDER REVIEW

Zeng, J., **Cheng, Z.**, Chen, H., Wang, Z., Thompson, J., Crosby, K. T., Han, H., Singhal, A., Ngo, W., Xia, C., Rosas-Rivera, D., Zhang, Z., Kang, M. H., Mao, Y., Diolaiti, M. E., Lee, G. C., Diffley, J. F. X., Song, Y., Qiu, L., Krah, N. M., Murthy, N., Jackson, R. N., Liu, Y., Ashworth, A., & Doudna, J. A. (2026). Targeting Cancer-Specific Mutations with RNA-Triggered Chromatin Shredding. *Nature*. <https://doi.org/10.1038/s41586-026-10738-7>